It's the Elicitation, Not the Eyes: Diagnosing VLM Failures in Slide Quality Inspection and Routing a Symbolic–Neural Critic

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Abstract

Vision–language models (VLMs) are the default critic inside slide- and document-generation agents, yet they routinely pass slides whose text visibly overflows its box. We ask whether this is a failure of *perception* (the model cannot see the defect) or of *elicitation* (it sees, but the way we ask buries the signal). We settle this per defect with an attribution protocol built on a lossless structured-geometry oracle, and find the answer is not uniform: some failures are sub-perceptual under this protocol, and no prompt recovers them; others are format-suppressed and reappear under targeted defect-specific elicitation with the model and image frozen; a further reference-assisted slice recovers only when a clean twin is available. The diagnosis prescribes the critic by routing each defect to the engine that owns its bottleneck, and a minimal symbolic–neural hybrid follows as a consequence rather than a design. On the nine image-level classes, that hybrid reaches mean balanced accuracy 0.86 against 0.59 for the conventional pointwise VLM-C0, with strict coverage reported cell by cell.

1 Introduction

An automatic critic that flags layout and content defects is now a load-bearing component of every slide- and document-generation agent, and the default critic is a general-purpose VLM asked to grade a rendered slide. These critics fail in a revealing way: shown a slide whose title visibly spills out of its box, a strong VLM under a standard rubric prompt answers "no defect." Whether this is a failure of *perception* (it cannot see the overflow) or of *elicitation* (it sees, but the way we ask buries the signal) is rarely asked, yet the answer decides whether the remedy is a larger model, a different prompt, or a different tool entirely. We treat that question as the design problem and answer it empirically.

A VLM that "cannot see" a slide defect is hiding two failures of opposite character, and separating them is the whole problem. Some defects are *sub-perceptual* (i.e., below the tested models' effective threshold under this protocol): a three-pixel alignment offset or a one-point font mismatch sits below the model's effective acuity, and no amount of prompting, scale, or resolution recovers it. Others are *format-suppressed*: the model can perceive them, but the conventional elicitation ("here is a slide and a rubric; score every defect type in one pointwise pass") buries the signal, while a targeted atomic query with required evidence recovers it with the model and image frozen. A third slice is *reference-assisted*: the model resolves the defect once a clean twin is available, but not from a single slide alone. Conflating these cases is why the field oscillates between "VLMs are bad at layout" and "VLMs can judge slides"; both hold, for different defects.

We make the split measurable with an attribution protocol built on a *lossless structured-geometry oracle*. Each slide carries an intermediate representation (IR) of element bounding boxes, text, and style; we present the same paired (defective, clean) slides under four input modalities (image-only, structured-oracle, VLM-caption, and image+oracle) crossed with detect/localize/repair tasks. Because the oracle is lossless machine-readable structure rather than a model-written caption, the perception channel it isolates is uncontaminated: a model that fails image-only but succeeds with the structured oracle is bottlenecked on perception, not reasoning. This is the methodological difference from concurrent perception/reasoning attribution that uses natural-language perception oracles, which re-inject the very perceptual error they try to factor out.

The diagnosis then prescribes the critic (Fig. 1): route each defect to the engine that owns its bottleneck. Sub-perceptual geometry goes to a symbolic linter that reads coordinates and rules; text and structural semantics go to a language model; and the format-suppressed, perceivable classes go to a VLM, but only under the changed elicitation the diagnosis shows recovers them. To stress-test the design we construct a render-containment overflow class that is invisible to the linter by construction (the declared box is legal; only the rendered pixels overflow) and that some specialized reward scorers miss while general perceptual scorers and a frontier VLM judge catch, so its detection tracks rendered-quality read-out rather than the training-domain label (Sec. 5.3). The full symbolic–neural critic is built for *IR-owning generation agents*, where the declared structure the linter relies on is available; for bare third-party pixels the deployable critic is the re-elicited VLM alone (Sec. 7), and coverage claims should be read within that scope.

Contributions. Our contributions form a chain. (i) A perception/reasoning attribution protocol built on a *lossless* structured-geometry oracle (element boxes, text, style—not a caption that re-injects perceptual error) that sepa-

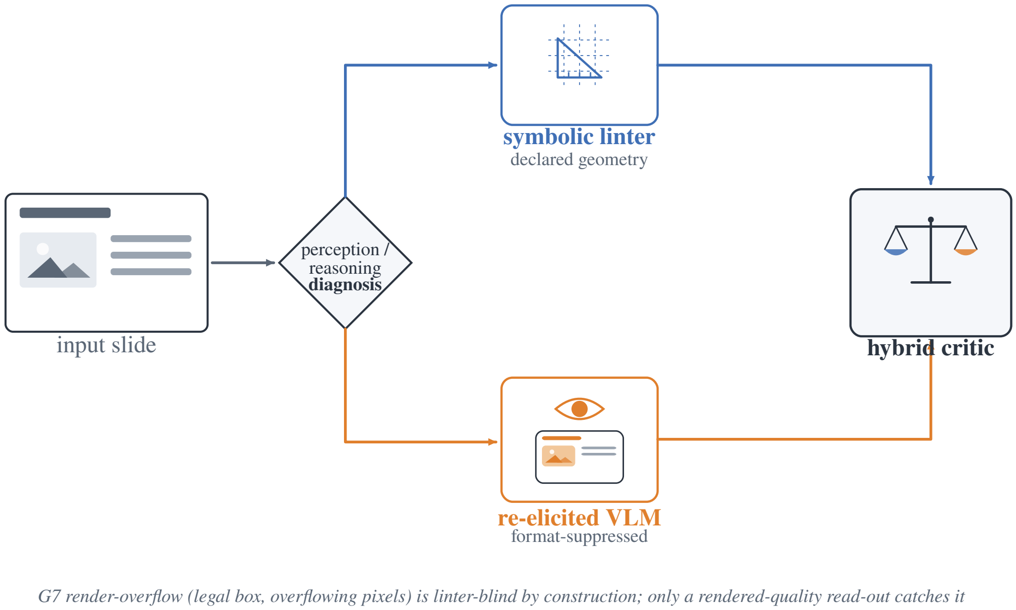

Figure 1: **The paper in one picture.** A perception/reasoning diagnosis (Sec. 4) shows slide-inspection failures are of two kinds. Declared-geometry defects are exact and cheap for a symbolic linter, with a sub-threshold acuity tail nothing recovers; merely format-suppressed defects are recovered by a vision–language model under a changed elicitation. Routing each defect to its bottleneck-appropriate engine yields a hybrid critic (Sec. 5). A constructed render-overflow class (legal declared box, overflowing pixels) is the crux: invisible to the linter by construction, it is caught only by engines with a rendered-quality read-out (a re-elicited VLM, a frontier VLM judge, or general perceptual reward models), not by domain-labelled scorers, so its detection tracks perceptual capability rather than training domain (Sec. 5.3).

rates VLM inspection failures into *sub-perceptual*, *format-suppressed*, and *reference-assisted* slices, reproducing on real CC-licensed layouts with no annotation (Secs. 4, 7). (ii) The central finding: targeted defect-specific elicitation recovers detection that the pointwise rubric buries, while G1/S6 remain a distinct reference-assisted case rather than evidence that the single-slide prompt "saw it all along" (Sec. 5). (iii) The diagnosis *prescribes* the critic: routing each defect to the engine owning its bottleneck (a linter for declared geometry, one re-elicited VLM for the rest) raises mean balanced accuracy to 0.86 from 0.59 for the conventional pointwise VLM-C0, while the linter-blind render class makes the blind spot falsifiable and a reward-side audit reproduces the split (Sec. 5). (iv) An 8B examiner finetuned on injected defects surpasses a zero-shot 30B in-distribution and abstains rather than hallucinates geometry, and a perturbation-fidelity audit shows many injected geometry defects never render, a hazard any IR-labeled dataset inherits (Sec. 6).

2 Related work

The VLM perception bottleneck. A growing line of work argues that VLM failures on visually grounded tasks are dominated by perception rather than reasoning, decomposing the bottleneck for chart understanding (Liu et al. 2025b) or showing that the needed visual information is present in the representation but unused (Fu et al. 2025); MMVP (Tong et al. 2024) is the canonical demonstration that VLMs stumble on minimally-different image pairs humans solve trivially, and its sub-perceptual acuity floor is what our title ("not the eyes") echoes. We share the perception-first stance but ask it of a different task family, *quality inspection of generated artifacts*, where the perception channel can be isolated exactly with structured geometry rather than approximated by captions, and where the answer feeds a critic architecture rather than a benchmark.

Perception/reasoning attribution. Concurrent work separates perception from reasoning in abstract visual-reasoning benchmarks via natural-language perception oracles (Wang et al. 2025). We share the goal but differ in kind: our oracle is a *lossless* structured representation rather than a lossy caption, a cleaner attribution boundary because a caption re-injects the very perceptual error it is meant to factor out; our domain of generated documents admits exact injected labels; and we close the loop from diagnosis to a deployed critic. Layout-error-detection work (Heo et al. 2025) introduces distribution-grounded structural-error injection but for *document-layout-analysis predictions*, and notes that existing metrics cannot isolate whether failures stem from recognition or reasoning, the gap our attribution protocol fills.

Relative vs. absolute judgement. Our finding that relative elicitation beats absolute scoring echoes work showing VLM judges can rank but cannot reliably score (Kumar et al. 2026); that result concerns a judge's *output mode*, orthogonal to our *source of failure*, and we adopt it to motivate the pairwise and atomic-binary heads of our critic. Comparative/relative elicitation is also known to recover discriminative signal that pointwise scoring suppresses in LLM/VLM judges (Liusie, Manakul, and Gales 2024), though the protocol choice is not bias-free and can itself distort preferences (Tripathi et al. 2025). Our contribution is not that effect but its *defect-class boundary*: it rescues text overflow (G1) and image/text contradiction (S6) yet not fine alignment (G3) or terminology drift (S3), which is what turns the elicitation effect into a perception/format attribution signal.

Slide generation, evaluation, and reward. Slide generation with design feedback is active: AutoPresent (Ge et al. 2025) treats it as tool-augmented program generation with self-refinement, aesthetic-reward systems (Pan et al. 2026; Liu et al. 2025a) report that iterative VLM refinement helps gross layout damage but fails on fine aesthetics and reward-hacks, and slide-evaluation studies (Kang, Bao, and Goswami 2025) call for calibrated critic-in-the-loop selection. These *motivate* a routed critic rather than compete: they observe the VLM's perceptual ceiling and route around it with rules, whereas we first *diagnose* that ceiling and show part of it is elicitation-recoverable. We probe real data with SlideAudit (Zhang et al. 2025) and audit five published scorers plus a closed frontier VLM-as-judge (Sec. 5.3). Defect injection adapts the paired-data critic recipe to rendered documents at zero annotation cost, and we use it as the semantic engine of the routed critic.

3 Problem setup and defect taxonomy

Task and representation. The critic inspects a single rendered slide, optionally with its declared structure, and must *flag and localize* defects of a fixed set of types; a detection counts only when it fires on a defective slide and stays silent on its matched clean twin. Every slide is an intermediate representation (IR) of typed elements with bounding boxes, text, and style; we inject defects programmatically into the IR and render, yielding exact labels and matched (defective, clean) pairs. The taxonomy (Table 1) has a *geometry* group G1–G6, a *semantic* group S1–S6, and one constructed render class G7; it is aligned to the expert-derived SlideAudit taxonomy (Zhang et al. 2025) via a bidirectional map (crosswalk in the supplement), and results are reported in its canonical classes. We add G7 because it is the sharpest test of the linter's blind spot: the declared box is legal (inside the safe margin, no overlap) but the rendered content overflows its container, so the defect exists *only* in pixels. On real agent-generated decks G7 is common, affecting most slides of a Zenodo10K slice and over half of an AutoPresent slice.[1]

[1] 93-deck Zenodo10K slice: 81/93 decks affected (857/3167 legal boxes); 14-deck AutoPresent slice: 8/14 decks (20/50 legal boxes).

Table 1: Defect taxonomy. The geometry group is detected symbolically; the semantic group is the examiner's domain; G7 is constructed to be linter-blind. "Channel" is the engine the diagnosis assigns (Sec. 4–5).

Group	Class	Description	Channel
Geometry	G1 text overflow	text exceeds its box	VLM+linter
	G2 element overlap	boxes intersect	linter
	G3 alignment offset	small mis-alignment	linter
	G4 font-size inconsistency	off-grid font size	linter
	G5 brand-color violation	off-palette color (ΔE)	linter
	G6 margin violation	element past safe margin	linter
Semantic	S1 title/body mismatch	body contradicts title	VLM
	S2 narrative-order break	deck sections out of order	LLM
	S3 terminology inconsistency	term drift across deck	term linter
	S4 density violation	slide over/under-packed	LLM
	S5 missing logic section	dropped argument step	LLM
	S6 image/text contradiction	figure contradicts caption	VLM
Render	G7 containment overflow	legal box, rendered overflow	VLM (atomic)

Attribution modalities and metric. To attribute a failure, we feed each paired sample to the model under four inputs: **A** the rendered image alone; **B** the lossless structured oracle (element boxes, text, style from the source IR); **B′** a VLM-written caption (a lossy control); and **C** both image and oracle. Under each we ask the model to detect, localize, and repair the defect. The attribution reads off the A-vs-B contrast: if the model fails on the image alone (A) but succeeds given the oracle (B), we call the class *perception-bottlenecked* operationally: the task becomes solvable when the same information is made explicit as lossless structure. This is a diagnosis of which engine can solve the class, not a claim about the model's internal representations. If it fails under both, the class is unsolved even at the structured interface, and we route it symbolically.[2] The headline metric is *balanced accuracy on the paired clean controls*: a detection counts only if it fires on the defective slide and stays silent on its matched clean twin, so a model cannot score by flagging everything.

Two linters. A geometry linter detects G1–G6 from coordinates and rules (brand color via CIELAB ΔE_{2000}, not RGB distance), at zero false positives on clean slides. A terminology linter handles S3 from an occurrence table plus near-duplicate clustering, image-free. Both are the "detectors of record" the diagnosis routes to.

4 Diagnosis: two kinds of blindness

Pointwise image-only inspection floors fine geometry. Under the conventional setup—one slide in, one rubric score out, no reference slide for comparison (we call this *pointwise image-only*)—G2–G6 sit at chance for 4B and 8B models, and a 30B breaks only G1 overflow. The floor is invariant to the obvious levers: swapping the vision encoder across *five* families (SigLIP2, an LLM-based encoder, InternViT, a native-resolution NaViT, MoonViT) leaves every model

[2] The oracle's ground-truth labels are stripped from the model's view by a regression-tested de-leaking pass, so B tests perception of structure, not label leakage. B′ is a control, not a second oracle: a caption re-injects the perceptual error it is meant to factor out, so B's advantage is by construction.

at the random line (differing only in abstain-vs-over-flag bias), and raising resolution 1536→2048 changes nothing. The lever is elicitation and magnitude, not encoder or pixel budget, so the floor is a property of the *pointwise* protocol, not a hard capability limit. Either way the right detector for declared geometry is the symbolic linter: it reads coordinates directly at balanced accuracy 0.75–1.0 with zero false positives, cheaper and exact where the IR is available.

But some apparent perceptual failures are elicitation artifacts. The same setting hides a second, opposite phenomenon. For G1 overflow and S6 image/text contradiction, pointwise balanced accuracy is 0.50, indistinguishable from the sub-threshold tail, yet a two-alternative forced choice between the defective and clean slide jumps to 1.00: the defect is perceivable, but the single-slide pointwise call does not surface it. We therefore treat G1/S6 as *reference-assisted*: recovery needs the clean twin, not just a renamed single-image question. G3 alignment behaves differently, but only *above a magnitude threshold*: sweeping the injected offset as an internal contrast (one element shifted from its aligned sibling column), the forced choice saturates to 1.00 by 16 px (0.83% of slide width) while the sub-threshold tail ($\leq$ 8 px) stays at chance even side by side. So fine alignment is not a class-level capability floor: above threshold it is format-suppressed and recoverable, with a sub-perceptual residue only in the small-offset tail. The connecting thread is a *magnitude-gated* recoverable-versus-sub-perceptual boundary rather than a clean geometric/semantic split, and S3 terminology (below) remains the one defect no elicitation rescues.

On the rebuilt corpus we also now have a human spot-check on the actual pairs cited here, not just the earlier diagnosis runs. Replacing only the stale legacy G3/G5 samples and keeping the other 55 pairs fixed, the v2 pass verifies every sampled defect as visible and every clean twin as clean (73/73, Wilson lower bound 0.95 for both), with source-structure faithfulness 55/55 on the auditable classes. The delta is exactly where the re-operationalisation says it should be: G3 shifts from 0/7 on the old external-reference sample to 8/8 on the corrected within-list misalignment sample, and G5 from 0/7 to 10/10 on the corrected within-list chromatic clash sample; the non-replaced 55 pairs are unchanged.

A counterexample: terminology. Not every consistency defect is format-suppressed. S3 terminology inconsistency is *not* rescued by forced choice (0.625, heavy position bias): reading the same term across deck pages and spotting a subtle variant is below threshold even side by side. The structured-oracle channel, which feeds the model the full deck text with both variants present, also fails (0.56): with both variants already explicit in the input, this failure cannot be OCR, and instead reflects cross-page term-matching that does not survive the yes/no framing. The image channel is consistent with fine-grained glyph discrimination and the oracle channel fails even with the text in hand, so neither route is reliable. S3 leaves the VLM entirely for a symbolic terminology linter, which reaches balanced accuracy 1.000 against the VLM's 0.69, a negative result that *re-routes* the class.

A magnitude-bounded geometric blind spot: G6 page offset. Re-posing a defect as a within-slide contrast does not rescue every class; on one it fails outright. We redefine G6 margin as a *page offset*: the whole content block is shifted toward one edge (every element together, so internal alignment is preserved and the defect cannot be misalignment), leaving one margin crowded against (or running off) the edge and the opposite side empty.[3] Across all six VLMs and both pointwise framings the page offset stays at chance (roster-mean C3 0.50), and the pairwise test that separates *data artifacts* (S6, which recovers to perfect once the figure is rendered) from genuine blindness leaves G6 on the floor for *every* model. A stress-test addendum identifies the boundary: when the same rigid offset moves content 48 px beyond the page edge, both C0 and C3 detect the defect at 0.958 balanced accuracy (11/12 positives, 12/12 clean controls), although C0 never names G6 while C3 does. Thus the blind spot is genuine over the tested moderate, unclipped range but not magnitude-invariant; overt clipping restores visibility. Declared geometry still routes to the linter because it detects both regimes without relying on this visual threshold.

A template pitfall that matters later. Enterprise "snap-to-master" templates re-fit element boxes to a grid before rendering, and we find a template silently *absorbs* 45% of injected geometry defects: overlap, alignment, and margin violations become pixel-identical to their clean twins after snapping, while overflow (content-determined) survives, a trap for any dataset that derives labels from the IR but evaluates on the rendered image, quantified and tied to a reward-model failure in Sec. 5.3.

5 From diagnosis to a symbolic–neural critic

The diagnosis says a single model is the wrong unit: route each defect to the engine that owns its bottleneck. We build that critic, validate the elicitation recovery it relies on across many models, and stress it with the constructed render class G7.

5.1 Elicitation: format suppression is not capability

Four ways to ask. Holding the model and image fixed, we compare four ways of prompting it. **C0** is the conventional baseline: one call that scores *every* defect type at once against a rubric. **C3** is the contrast: one call per defect type, asking only a yes/no question and requiring the model to point at the offending element. **C1** (free-form describe-then-classify) and **C2** (a synthetic-twin pairwise, where the clean reference is re-rendered to a grid) are two further variants we report for completeness. The scientific test is C3 vs. C0—same model, same image, same taxonomy, differing only in "ask everything at once" versus "one yes/no per type with forced evidence." If C3 $\gg$ C0, the VLM's silence under C0 was a format artifact, not a capability gap.

[3] The original S6 render omitted the figure element, so the cross-modal contradiction was literally unseeable (recall 0); on the valid-figure corpus the S6 2-AFC reproduces at 0.75–1.00 across all six VLMs, confirming it is a render artifact, not blindness.

C3 recovers the linter-blind class across the capable subset. We scope the claim to models that can see G7 at all: a model counts as "capable" only if, under C3, it both detects the overflow (balanced accuracy ≥ 0.70 and precision ≥ 0.70 on paired clean controls) and points at the overflowing element; selected on the same test items, this subset is descriptive rather than confirmatory. This excludes InternVL-8B and Ovis-9B, which stay near chance, a capability boundary rather than a format effect. On the four capable models (three scales, three vendors), C3 recovers what C0 suppresses: balanced accuracy 0.50–$0.79 \rightarrow 0.93$–1.00, precision 0.88–1.00. The effect is cross-model: pairing each slide across the four models, C3 fixes 233 G7 items that C0 missed against 2 reversals (χ^2_1=227, $p < 10^{-50}$).[4] The mechanism is simple: C0's one-call rubric has no slot for G7 (it is a render class, not in the original taxonomy), so the model has no name to attach to what it sees and abstains; C3 asks the question directly.

Targeted elicitation recovers what the rubric suppresses. Is it naming the class that helps, or changing how we ask? Two controls separate this. *C0+* is C0 with G7 simply added to the rubric's defect list—same one-call format, now the model has a name for the class. *C0_named* instead asks about G7 alone, on its own yes/no call. Table 2 shows the split: adding the name to the one-call rubric (C0+) recovers *recall* (234 defectives newly flagged, p=7×10^{-71}) but wrecks *specificity*—it then over-flags clean slides on 55% of pairs versus 2.4% for C3, so balanced accuracy stalls. Asking about G7 on its own call (C0_named) recovers both, and supplying a clean reference slide adds nothing on top. So for G7 the suppressor is the *one-call format*: under a targeted atomic query, the same model on the same image recovers the detection the whole-taxonomy rubric buries. The boundary is G1/S6: naming alone does *not* help—those need the clean reference slide (pairwise choice $\rightarrow$ 0.97–1.00), a *reference-assisted* effect rather than single-slide format suppression. A clean-vs-clean control rules out a forced-pick artifact (capable models answer "tie" on 92–100% of clean pairs).

Compute-matched control: recovery without additional test-time compute. The remaining alternative is that C3 wins only because it spends more test-time compute than the one-call rubric. We test that directly on three API-served models spanning distinct vendors, holding the image and rendered corpus fixed and matching C3's budget with repeated C0 draws. The headline control is *self-consistency*: majority vote over K=10 independent C0 samples. On G7, C3 still beats that compute-matched control on all three vendors ($0.92/0.88/0.83$ vs. $0.61/0.59/0.63$; $\Delta = +0.31/+0.29/+0.20$), and all three contrasts survive Holm correction within the full 24-test family (reported in the supplement). The budget points the same way: C3 is the cheapest condition in completion tokens ($34.7/37.5/40.5$ output tok per slide, one call) while the matched self-consistency baseline spends 8–10 calls per slide and still loses. The result is therefore not that a heavier budget rescues the rubric; it is that the whole-taxonomy pointwise format suppresses detection, and breaking that format recovers it without additional test-time compute. We report the union aggregation of the same draws, the definition-matched one-call control, and the G1 negative control in the supplement.

[4] Per-model $p \leq 1.5{\times}10^{-11}$, all surviving Holm correction over the pre-registered 85-test family; the two reversals are on Gemma4-31B (C3 recall 88/90). Full table in the supplement.

Table 2: Naming vs. format on G7. Adding the class name to the one-call rubric (C0+) recovers recall but wrecks specificity (55% clean over-flag); asking about G7 on its own call (C0_named, C3) recovers both. C0_named matches C3, so a clean reference adds nothing for G7.

elicitation (G7)	clean over-flag	bal-acc
C0 (whole-tax., unnamed)	2.4%	0.50
C0+ (whole-tax., G7 named)	55%	0.58–0.82
C0_named (atomic, no ref.)	2.4%	**0.93–1.00**
C3 (atomic, +forced loc.)	2.4%	**0.93–1.00**

No universal elicitation, so the critic routes. The best elicitation is jointly set by model capability and defect class—there is no one prompt, which is why the critic routes per defect and model: the C2 synthetic-twin pairwise rescues declared G1 on capable models (snapping absorbs the overflow and the contrast exposes it), while C1 free-form rescues weak models but collapses on strong ones (precision ≈ 0.50). The format effect is not a synthetic artifact: on human-annotated SlideAudit (no structure) C3 still beats C0 for every class (e.g. brand color, alignment). Absolute geometry on bare real pixels stays modest outside blatant overlap—real fine geometry needs the linter, hence the structure bare images lack, so on image-only real data the hybrid degrades to VLM-only, a ceiling set by missing structure, not method.

5.2 The minimal hybrid critic covers the most

Complementary coverage the diagnosis predicts. On a separate novel-template held-out set (150 defective/clean pairs per class), the corrected frozen route reaches macro balanced accuracy .957 and covers 8/9 classes; the sole miss is S6 (.677 balanced accuracy, .607 precision), whose perfect recall is offset by clean false positives (full table and fidelity audit in the Technical Supplement). We score each critic by *named attribution*: a detection is correct only if it names the right defect type on the defective slide and stays silent on the matched clean slide. We call a class "covered" when this balanced accuracy exceeds 0.75 (with the 95% Wilson lower bound above 0.65, so the cell is not just lucky). Coverage tracks the diagnosis: declared geometry is covered by the linter, the format-suppressed classes by a C3-elicited VLM. The headline is therefore the *mean balanced accuracy* of the combined critic: the minimal hybrid of linter plus one C3-elicited VLM reaches 0.86, against 0.59 for the conventional pointwise VLM-C0 and 0.66 for the linter alone (Table 3). Under the same strict covered criterion, the minimal hybrid clears 6 of the 9 image-level classes; G6 at 0.75 $_{[.67,.80]}$ misses by the paper's own "exceeds" rule, and S1 under linter+C3 misses because its lower bound stays below 0.65. The nine classes are the image-level ones; G4/S3 are linter-owned and

Table 3: Per-class named-attribution balanced accuracy [95% Wilson]; n = paired positives (G1 40, G2 40, G3 70, G5 40, G6 80, G7 40, S1 18, S4 36, S6 12; precision per cell in the Technical Supplement). The minimal linter+VLM-C3 hybrid has the highest mean balanced accuracy. "covered" means bal-acc > 0.75 and Wilson lower bound > 0.65; under that rule, linter+VLM-C3 and the corrected routed hybrid each clear 6/9 classes. The routed column reports the corrected S1 route; the frozen text-only S1 route (0.25, precision 0.09) is reported in the prose below. Small S1/S6 cells are routing diagnostics, not confirmatory (those claims rest on the elicitation decomposition and 2-AFC).

Class	linter	VLM-C0	VLM-C3	linter+C3	routed
G1 overflow	1.00 [.91,1]	0.50 [.46,.54]	0.50 [.46,.54]	**1.00 [.91,1]**	**1.00 [.91,1]**
G2 overlap	0.80 [.68,.87]	0.71 [.60,.79]	**0.86 [.74,.92]**	0.80 [.68,.87]	0.80 [.68,.87]
G3 alignment	0.93 [.85,.96]	0.62 [.55,.68]	0.71 [.63,.77]	**0.93 [.85,.96]**	**0.93 [.85,.96]**
G5 color	1.00 [.91,1]	0.50 [.46,.54]	0.68 [.57,.75]	**1.00 [.91,1]**	**1.00 [.91,1]**
G6 margin	0.75 [.67,.80]	0.51 [.40,.61]	0.51 [.47,.55]	**0.75 [.67,.80]**	**0.75 [.67,.80]**
G7 render-overflow	0.00 [0,.54]	0.50 [.46,.54]	**1.00 [.91,1]**	**1.00 [.91,1]**	**1.00 [.91,1]**
S1 title/body	0.50 [.41,.59]	**0.94 [.75,.98]**	0.83 [.63,.92]	0.83 [.63,.92]	**0.94 [.75,.98]**
S4 density	0.50 [.45,.55]	0.51 [.44,.59]	**0.93 [.81,.97]**	**0.93 [.81,.97]**	0.72 [.60,.80]
S6 image/text	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]
mean / covered	0.66 / 4/9	0.59 / 1/9	0.72 / 3/9	**0.86 / 6/9**	0.85 / 6/9

S2/S5 are deck-scope (Sec. 6). Note that this routing needs no per-class tuning: the rule follows from the diagnosis, not from optimizing over class outcomes, so the coverage table is what the diagnosis predicts rather than a tuned architecture.

A data-driven routing correction. We initially routed S1 title/body mismatch to the text LLM, but the text-only probe over-flags (0.25, precision 0.09) while the image-bearing VLM names it reliably (0.94, precision 0.90): title/body mismatch needs the *rendered layout*, not just the text. We therefore report both numbers: the frozen route as the pre-registered baseline, and the corrected VLM route as the deployed fix. Under the corrected route, S1 uses the VLM under its open pointwise prompt (C0, 0.94) rather than C3 (0.83), since recall is already saturated and C3 only adds false positives—a concrete case of per-class elicitation routing. The one image-level class the hybrid does not cover is S6 image/text contradiction, at 0.50 single-image for every engine; the pairwise test shows the model *does* perceive it (recall 1.0, specificity $\sim$0, over-asserting rather than missing), so this is a reference-assisted effect, not a blind spot, in contrast to the moderate, unclipped G6 page-offset regime.

5.3 G7: the linter-blind class that specialised rewards also miss

The construction and split. G7 makes the linter's blind spot falsifiable: a declared box that is legal (in-margin, non-overlapping) but whose rendered content overflows its container. The IR carries only legal short text; the overflow lives in the renderer, so the linter and any structure-only model are blind by construction—a self-check confirms zero findings on 90/90 G7 defectives. We then audit five published scorers plus one frontier VLM judge on the same paired slides. The decisive result is a same-backbone dissociation: CLIP-IQA detects G7 while the LAION aesthetic head on the *same* CLIP-L/14 is at chance; Skywork, PickScore, and Qwen3.7-Max also detect it, whereas DocReward does not (full per-scorer table in the Technical Supplement). So on the scorers tested, G7 tracks a rendered-quality read-out rather than the training-domain label.

Perturbation fidelity: 45% of injected geometry never renders. The reward audit ties to the template result of Sec. 4. Rendering every injected defective both faithfully and snapped-to-master, 45% of injected geometry defects (100% of overlap, alignment, margin; 20% of overflow) become pixel-identical to their clean twin after snapping. Feeding the *snapped* pairs to all five scorers gives preference accuracy 0.00 at a reward gap of exactly 0.000: identical inputs force identical scores. This is a dataset-integrity hazard, not a reward-model deficiency, and it is model-agnostic by construction: any reward model trained or evaluated on snap-rendered images with IR-derived labels inherits these zero-signal pairs. Perturbation fidelity must be checked at the pixel level, not assumed from the IR.

6 The semantic engine: a specialist examiner

The hybrid's semantic engine can be a zero-shot model, but a small specialist does better on its own ground. We finetune Qwen3-VL-8B with QLoRA on 2,142 injected supervision records; the training mix is summarized in the Technical Supplement.

An 8B examiner surpasses a zero-shot 30B on *in-distribution* semantics. On in-domain held-out page-semantic inspection, the finetuned 8B reaches balanced accuracy 1.000 on both S1 title/body and S4 density, beating both zero-shot baselines including the 30B. The headline is simple: on the examiner's native in-distribution semantic tasks, the specialist 8B beats the larger zero-shot model. The largest gap is S4 density (1.00 vs. 0.77 for the 30B), and both in-distribution contrasts survive Holm correction over the 85-test family. Full format-control and severity breakdowns are in the Technical Supplement.

It abstains instead of hallucinating geometry. Under image-only inspection, the finetuned model stays at 0.50 on G2–G6 with *zero* false positives: trained with image-only abstention targets, it declines to hallucinate geometry from pixels and leaves those classes to the linter, the designed behavior, and the reason the hybrid's geometry column comes from the linter rather than the VLM.

The main negative result is sim-to-real. The examiner does not transfer as a bare-pixel real-world inspector. On image-only SlideAudit, its synthetic S4 strength collapses to chance (0.50), whereas only the zero-shot 30B shows real density signal (p=1.1×10^{-5}). Scale, not finetuning, carries bare-pixel transfer: the examiner learned a distribution, not universal slide quality.

7 External validity

A real pre-sales agent deck. On a real 16-slide proposal from a deployed PowerPoint agent, the verifiable defect is

20 unfilled template placeholders ("add a title here") across 7 slides—a semantic-completeness defect the linter is blind to. The zero-shot 30B is the best detector (recall 1.0, names the placeholder on 4/7 slides, scores them 0.03 vs. 0.64 for clean), while the *finetuned* 8B, top of the intrinsic ranking, is among the *worst* here (0/7), and the linter's apparent recall is spurious geometry noise: on a novel defect the ranking is by general-VLM ability, not specialist in-distribution quality—the examiner learned a distribution, not universal quality. Feeding the flagged placeholders back to the generator for one revision drives the defect from 20 to 0.

The structured attribution holds on real layouts. The body runs the A/B/C attribution on synthetic slides; the cleanest discrimination needs it on *real* geometry with a *real* oracle. We obtain it with no human or model annotation: ZENODO10K (Zheng et al. 2025) ships real CC-licensed `.pptx` with intact element XML, so `python-pptx` extracts a geometry/text/style oracle *lossless with respect to the declared IR* (the IR$\rightarrow$render gap that G7 exploits is exactly what this oracle does *not* capture). From 26 real decks we inject one single-shape defect per slide *in PPTX XML space* (overflow, overlap, alignment, font, margin) and render both the clean and defective one-slide deck with the same renderer (LibreOffice), so the paired pixel diff isolates the injected defect on real pixels (209 paired slides, 5 classes, 3 VLMs / 3 families). Two results follow. First, the §4 "snap absorbs 45%" hazard is *template-specific*: real free-form decks absorb only 7% (render-fidelity 0.93), a positive control. Second, *all three* attribution outcomes appear on real geometry: coarse defects (overflow, overlap) are image-sufficient (A 0.70–0.74, structure adds nothing); a margin bleed is a perception bottleneck the oracle rescues for weaker VLMs (A $0.59 \rightarrow$ B 0.70); and fine alignment stays near chance in *every* channel for *every* model (A=B=C$\approx$ 0.50; a supra-threshold $44\,$px internal-contrast re-run reproduces this, A/B/C 0.51–0.54, n=41). This is *not* a hard capability floor (on clean slides the same defect recovers under targeted elicitation once it clears acuity), but on real cluttered layouts the misaligned element cannot be isolated among many genuine ones and specificity collapses, so for declared geometry the symbolic linter remains the reliable detector regardless.

Open-world: structure recovered from pixels does not restore the linter. A third-party PDF/PNG deck ships no IR, so we ask whether *recovering* element structure from pixels re-arms the linter. Following DocLayNet/PubLayNet (Pfitzmann et al. 2022; Zhong, Tang, and Yepes 2019), PP-DocLayoutV2 returns region boxes per render (class-agnostic NMS removes cross-class duplicates) that feed the *same* linter. The answer is a pre-registered negative: recovered structure leaves the linter at chance on G1/G3/G4 and rescues only overlap partially (G2 0.54 vs. native-IR 0.67 vs. VLM-only 0.87); the detector *over-segments* ($\approx$ $2\times$ more boxes, recovery recall at IoU$\geq$$0.5$ only 0.30–0.39), so collisions split and overlap recall falls $0.78 \rightarrow 0.29$. On image-only SLIDEAUDIT the recovered linter clears chance only on G2 (0.55) while the C3-elicited VLM reaches 0.94. On real third-party decks even *native* IR is weakened (no position/font bookkeeping, so G3/G4 rules are at chance and margin/overlap rules *over-fire*, specificity $0.11/0.56$), so the VLM-only engine *dominates every coarse class*: the linter's edge is an in-distribution, clean-native-IR phenomenon, and open-world the deployable critic is VLM-bound.

8 Limitations

We now report a human spot-check on the rebuilt corpus, but not an inter-annotator or deployment-time human baseline. On the rebuilt v2 sample, every sampled defect is judged visible and every clean twin clean (73/73; 55/55 source-faithful on the auditable structure-carrying subset), so the paper's use of G3/G5 rests on corrected human-verified samples rather than the stale legacy framing. Three scope bounds remain. First, the real-layout attribution (§7) uses *injected* defects, so image-only SLIDEAUDIT is the complementary real probe. Second, the reward-audit document-reward miss is instance-level, so the categorical read rests on the two general-multimodal backbones and the same-backbone CLIP-IQA/LAION dissociation. Third, the examiner$\rightarrow$generation downstream effect is near-null: quality stays at Δq=$+0.00$ where a coverage critique moves it by $+0.32$. The contribution is the *diagnosis* and routing, not a claim to match human inspectors.

9 Conclusion

A slide-defect critic should not be one model. Telling apart sub-perceptual, format-suppressed, and reference-assisted failures turns "VLMs are bad at layout" into a routing rule: declared geometry to a symbolic linter, targeted single-slide elicitation for the format-suppressed classes, and paired reference use where a clean twin is what resolves the defect. A linter plus one re-elicited VLM reaches mean balanced accuracy 0.86 against the conventional VLM-C0's 0.59, with 6/9 image-level classes strictly covered. A constructed render-overflow class makes the linter's blind spot falsifiable, and the contribution is the diagnosis from which the routed critic follows. *Reproducibility:* code, linters, harness, and reward adapters are in the Code and Data Supplement; family-size updates are reported in the supplement.

References

Fu, S.; Bonnen, T.; Guillory, D.; and Darrell, T. 2025. Hidden in Plain Sight: VLMs Overlook Their Visual Representations. arXiv:2506.08008.

Ge, J.; Zhao, Y.; He, J.; Sun, Y.; Lu, Y.; Wu, Y.; Zhang, S.; Wei, J.; and Zhang, L. 2025. AutoPresent: Designing Structured Visuals from Scratch. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Heo, I.; Hwang, T.; Jung, J.; and Jung, S. 2025. LED Benchmark: Diagnosing Structural Layout Errors for Document Layout Analysis. arXiv:2507.23295.

Kang, H.; Bao, E.; and Goswami, A. 2025. VLM-SlideEval: Evaluating VLMs on Structured Comprehension and Perturbation Sensitivity in PPT. arXiv:2510.22045.

Kumar, D.; Tayebati, S.; Naik, D.; Krishnan, R.; and Trivedi, A. R. 2026. VLM Judges Can Rank but Cannot

Score: Task-Dependent Uncertainty in Multimodal Evaluation. arXiv:2604.25235.

Liu, C.; Yang, Y.; Zhou, K.; Zhang, Z.; Fan, Y.; Xie, Y.; Qi, P.; and Wang, X. E. 2025a. Presenting a Paper is an Art: Self-Improvement Aesthetic Agents for Academic Presentations (EvoPresent/PresAesth). arXiv:2510.05571.

Liu, J.; Zeng, W.; Zhang, X.; Wang, Y.; Shan, Z.; and He, J. 2025b. On the Perception Bottleneck of VLMs for Chart Understanding. arXiv:2503.18435.

Liusie, A.; Manakul, P.; and Gales, M. 2024. LLM Comparative Assessment: Zero-shot NLG Evaluation through Pairwise Comparisons using Large Language Models. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.

Pan, Y.; Hu, C.; Huang, X.; Huang, C.; Zhao, M.; Bi, Y.; Zhang, X.; Zeng, A.; and Hu, L. 2026. AeSlides: Incentivizing Aesthetic Layout in LLM-Based Slide Generation via Verifiable Rewards. arXiv:2604.22840.

Pfitzmann, B.; Auer, C.; Dolfi, M.; Nassar, A. S.; and Staar, P. W. J. 2022. DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.

Tong, S.; Zhu, Y.; Wu, P.; Li, X.; Wu, Y.; Wang, M.; Zhang, Y.; Li, Z.; He, Y.; Sun, Y.; Xie, T.; Li, Y.; Zhang, Y.; Snavely, N.; LeCun, Y.; and Li, S. 2024. Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Tripathi, S.; Wadhwa, G.; Durrett, G.; and Niekum, S. 2025. Pairwise or Pointwise? Evaluating Feedback Protocols for Bias in LLM-Based Evaluation. arXiv:2504.14716.

Wang, X.; Huang, J.; Zhang, X.; Wang, T.; and Ma, J. W. 2025. Your Reasoning Benchmark May Not Test Reasoning: Revealing Perception Bottleneck in Abstract Reasoning Benchmarks. arXiv:2512.21329.

Zhang, Z. J.; Chen, R.; Zhong, M.; and Wobbrock, J. O. 2025. SlideAudit: A Dataset and Taxonomy for Automated Evaluation of Presentation Slides. arXiv:2508.03630.

Zheng, H.; Guan, X.; Kong, H.; Zheng, J.; Zhou, W.; Lin, H.; Lu, Y.; He, B.; Han, X.; and Sun, L. 2025. PPTAgent: Generating and Evaluating Presentations Beyond Text-to-Slides. Introduces the Zenodo10K corpus of real `.pptx`; dataset `Forceless/Zenodo10K`, arXiv:2501.03936.

Zhong, X.; Tang, J.; and Yepes, A. J. 2019. PubLayNet: Largest Dataset Ever for Document Layout Analysis. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR)*.